

[ ENGLISH VERSION ]

(For Regular & External Candidates)

Time—Three Hours Fifteen Minutes

(First FIFTEEN minutes for reading the question paper)

Full Marks { 90 – For Regular Candidates  
100 – For External Candidates



Only the External Candidates will answer Group 'E'.

Figures in the margin indicate full marks for each question.

GROUP—A

1. Multiple choice questions. Four alternative answers are given below each of the following questions. Write the correct one:  $1 \times 15 = 15$

1.1 Which of the following rays has the highest capacity to ionise gases ?

- (a)  $\alpha$ -ray (b)  $\gamma$ -ray  
(c)  $\beta$ -ray (d) light ray

1.2 The characteristics of a fuse wire are :

- (a) high resistance, low melting point  
(b) low resistance, high melting point  
(c) low resistance, low melting point  
(d) high resistance, high melting point



1.3 How many groups are there in modern long form of periodic table ?

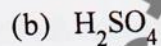
- (a) 7 (b) 17  
(c) 15 (d) 18



1.4 Which of the following compounds has no existence as separate and distinct molecule ?



1.5 Which acid is a weak electrolyte ?



1.6 What colour is produced when  $\text{H}_2\text{S}$  gas is passed through a solution of potassium dichromate acidified with dilute Sulfuric acid ?

(a) Orange

(b) Violet

(c) Green

(d) Deep blue

1.7 Which of the following is an ore of Aluminium ?

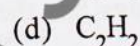
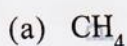
(a) Hematite

(b) Bauxite

(c) Malachite

(d) Chalcocite

1.8 Identify the triple bond unsaturated hydrocarbon :



1.9 Identify the greenhouse gas :

(a) Oxygen

(b) Hydrogen

(c) Water vapour

(d) Nitrogen

1.10 Volume of 22 g of  $\text{CO}_2$  at S.T.P. is  $[\text{C} = 12, \text{O} = 16]$  :

(a) 22.4 litre

(b) 11.2 litre

(c) 2.24 litre

(d) 1.12 litre

1.11 Find how many g of calcium oxide which will be obtained by intensely heating 10 g of calcium carbonate. Consider that calcium carbonate has completely decomposed.

$[\text{Ca} = 40, \text{C} = 12, \text{O} = 16]$

(a) 4.4 g

(b) 5.6 g

(c) 10 g

(d) 100 g



1.12 The coefficient of thermal conduction depends on :



- (a) the difference in temperature between the two ends of the conductor
- (b) property of the material of the conductor
- (c) the length of the conductor
- (d) the area of cross-section of the conductor

1.13 Which parameter remains unaltered during refraction of light ?

- (a) Velocity
- (b) Amplitude
- (c) Frequency
- (d) Wavelength



1.14 Which type of mirror produces virtual, diminished and erect image ?

- (a) Plane mirror
- (b) Concave mirror
- (c) Convex mirror
- (d) Parabolic mirror

1.15 Which one of the following is the unit of electrical energy ?

- (a) Watt
- (b) Ohm
- (c) Kilowatt-hour
- (d) Volt

#### GROUP—B

2. Answer the following questions (alternatives are to be noted) :

2.1 State the S.I. unit of radioactivity.



1

OR

What change will occur in atomic number if one  $\beta$ -particle is emitted from  ${}_{92}^{235}\text{U}$  ?

1

2.2 Match the Right column with the Left column :

1×4=4

| Left Column                                                   | Right Column |
|---------------------------------------------------------------|--------------|
| 2.2.1 Metal present in German silver                          | (a) Li       |
| 2.2.2 A transuranium element                                  | (b) Zn       |
| 2.2.3 The metal extracted from calamine                       | (c) Pu       |
| 2.2.4 Alkali metal present in second period of periodic table | (d) Ni       |



2.3 Write whether the following statement is true or false :  
Electrolysis always involves the process of oxidation-reduction.  1

2.4 Which gas is produced at the cathode during electrolysis of acidified water ? 1

OR

Which of the following is/are electrolyte(s) — Sugar solution in water, Ethanol and Acetic Acid ? 1

2.5 What happens when an iron spoon is dipped into acidified copper sulphate solution ? 1

2.6 Which metal is used as catalyst in Haber's process for preparation of ammonia ? 1

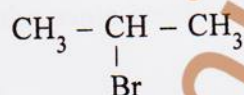
OR

Fill up the blank :



2.7 Name the metal which is extracted in thermite process. 1

2.8 Write down the IUPAC name of the following organic compound : 1



OR



Give an example of positional isomerism. 1

2.9 Write the name of monomer of a polymer which is used to prepare non-stick utensils. 1

2.10 Which layer of atmosphere is susceptible to storm and rain ? 1

OR

What is the unit of concentration of ozone layer ? 1

2.11 Name the greenhouse gas discharged from a refrigerator. 1

2.12 Write whether the following statement is true or false : 1

According to Charles' law, at  $(-273^\circ\text{C})$ , the volume of any gas is infinite.

2.13 What is the number of molecules present in 16 g of  $\text{O}_2$  at S.T.P. ?  1



2.14 Write whether the following statement is true or false : 1

If coefficient of linear expansion of iron is  $1.2 \times 10^{-5}/^{\circ}\text{C}$  then coefficient of cubical expansion of iron is  $3.6 \times 10^{-5}/^{\circ}\text{C}$ .

OR

Which physical quantity has the unit  $\text{Wm}^{-1}\text{K}^{-1}$  ? 1

2.15 What type of lens is used as a magnifying glass ? 1

2.16 For which value of the angle of incidence in refraction of light is Snell's law not applicable ? 1

2.17 What is now used in electrical circuits instead of fuse wire ? 1

2.18 State the resistance of 220 V - 100 W bulb. 1

GROUP—C

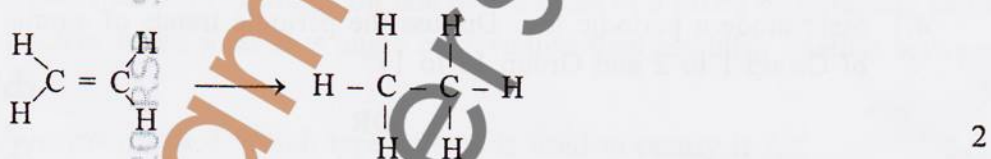
3. Answer the following questions (alternatives are to be noted) :

3.1 Name two copper containing alloys. State one use of each alloy. 1+1

OR

Explain the difference between minerals and ores. 2

3.2 How can the following conversion be carried out ?



OR

Write the names of two bio-degradable polymers. 2

3.3 Explain why the ozone layer is called 'natural sun screen'. 2

OR

How are global warming and greenhouse effect related ? 2

3.4 The volume of a certain amount of gas at a temperature of  $27^{\circ}\text{C}$  and at pressure 76 cm of Hg is 200 c.c. Find the volume of the gas when pressure of the gas changes to 38 cm of Hg and temperature changes to  $127^{\circ}\text{C}$ . 2

OR

Find the volume of 14 g Nitrogen gas at a temperature of  $227^{\circ}\text{C}$  and at pressure of 83.14 cm of Hg [ $R = 8.314 \text{ J mole}^{-1} \text{ K}^{-1}$ ]. 2

- 3.5 Explain with a suitable ray diagram the process of formation of image by concave lens. 2

OR

Explain with a diagram why the sky appears blue. 2

- 3.6 A thin wire of resistance  $4\ \Omega$  is bent to form a circle. Find the resistance across any diameter. 2

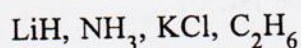
- 3.7 Draw electron-dot structure of calcium oxide. 2

[Atomic numbers of Ca and O are 20 and 8 respectively]

OR

Explain why the melting point of sodium chloride is much greater than that of glucose. 2

- 3.8 Identify the electrovalent and covalent compounds from among the following compounds : 2



- 3.9 Write with balanced chemical equation, what happens when dry ammonia gas is passed over heated sodium. 2

GROUP—D

4. Answer the following questions (alternatives are to be noted) :

- 4.1 State modern periodic law. Discuss the periodic trends of atomic radii for elements of Group 1 to 2 and Group 13 to 17. 1+2

OR

Explain the position of hydrogen in the modern periodic table. Which group of the periodic table contains solid, liquid and gaseous elements ? 2+1

- 4.2 What materials are used as cathode, anode and electrolyte during electroplating of a brass spoon with nickel ? 3

- 4.3 Write with balanced chemical equation, what happens when hydrogen sulphide is passed through aqueous solution of lead nitrate. 3

- 4.4 Write the structural formulae of 1, 2-dibromoethane and 1, 1, 2, 2-tetrabromoethane. Write the name of the organic compound which is produced when ethyl alcohol is heated with concentrated sulphuric acid. 2+1

OR

Mention one use of CNG. Discuss one each of the harmful effects of methanol and ethanol. 1+2



4.5 State Avogadro's law. Use simple calculation to show that moist air is lighter than dry air. 1+2

4.6 In a closed vessel 1 g of magnesium is burnt with 0.5 g of oxygen gas. Which reactant remains in excess ? Find the quantity of excess reactant. [Mg = 24, O = 16]. 3

OR



How many grams of  $\text{CaCO}_3$  will react with excess of dilute HCl to produce 66 g of  $\text{CO}_2$  ? [Ca = 40, C = 12, O = 16]. 3

4.7 Why are there gaps at regular interval between consecutive rails in rail track ? State the relation between linear expansion coefficient ( $\alpha$ ), superficial expansion coefficient ( $\beta$ ) and volume expansion coefficient ( $\gamma$ ). 2+1

OR



Find the length of an iron rod at  $110^\circ\text{C}$  if at  $10^\circ\text{C}$ , length of the rod is 20 cm. [volume expansion coefficient of iron =  $36 \times 10^{-6}/^\circ\text{C}$ ]. 3

4.8 Prove that with a rectangular glass slab, incident ray and emergent ray are parallel to each other. 3

OR



A ray of light is incident normally on one of the faces of a prism of refracting angle A and refractive index  $\mu$ . If D = angle of deviation then establish relation between  $\mu$ , A and D. 3

4.9 What is Hypermetropia ? Which type of lens is used to rectify it ? 2+1

4.10 The series combination of three 20 ohm resistances is connected in parallel combination to a 30 ohm resistance. Determine the equivalent resistance of the final combination. 3

OR

Three equal resistors connected in series across a source of e.m.f. together dissipate 10 W of power. What would be the power dissipated if the same resistors were connected in parallel across the same source of e.m.f. ? 3

4.11 State Lenz's law. "Lenz's law actually follows the principle of conservation of energy"—Justify. 1+2

4.12 What do you mean by mass defect ? Actual mass of proton, neutron and helium nucleus are 1.00728, 1.00867 and 4.0015 amu. What is the mass defect of ( ${}^4_2\text{He}$ ) helium nucleus ? 1+2





GROUP—E

(For External Candidates Only)

1×4=4

5. Answer the following questions (any **four**) :

5.1 Which gaseous hydrocarbon is used for ripening of fruits ?

5.2 State one harmful effect of ultraviolet rays.

5.3 What is the S.I. unit of specific resistance or resistivity ?

5.4 Write the volume of one mole oxygen gas at S.T.P.

5.5 Which one among the radioactive rays is an electromagnetic wave ?

2×3=6

6. Answer the following questions (any **three**) :

6.1  $\text{CH}_3\text{COOH}$  is an organic compound but  $\text{NaHCO}_3$  is not an organic compound – give reason.

6.2 State Fleming's left hand rule.

6.3 Why is the red colour selected for danger signal lights ?

6.4 Give one example of reducing property of  $\text{H}_2\text{S}$ .